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Venture-Backed Firms: How Does Venture Capital Involvement Affect Their Likelihood of Going Public or Being Acquired?

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This paper investigates how venture capitalists' involvement in new ventures affects the likelihood of entrepreneurial exit, either via an acquisition or via an initial public offering. We examine the prominence of venture capitals (VCs), the number of VCs invested in a company, as well as the timing, duration, and magnitude of their investments in new ventures. We find that each of these dimensions directly explains entrepreneurial exit, although their effects tend to differ depending on whether the exit occurs via an acquisition or an initial public offering (IPO). These results withstand several robustness checks and offer a more precise account of how the relationship between new ventures and VC firms unfolds in the early years of the entrepreneurial cycle.

Introduction

One of the most sought-after milestones for entrepreneurs is obtaining venture capital (VC) support. Not only do VCs provide financial backing to entrepreneurs, who often lack the necessary skills to deal with potential investors (e.g., Arikan & Capron, 2010; Leiblein & Reuer, 2004), but they also help recruit talented managers, formulate new strategies, and use their networks to garner resources for the company (Gompers & Lerner, 2004). In addition to offering these direct benefits, the backing of venture capitalist firms can also send a powerful signal regarding the quality of a new venture (Brav & Gompers, 1997; Carter & Manaster, 1990; Stuart, Hoang, & Hybels, 1999)—since VCs invest in only about 1% of the proposals they receive (Megginson & Weiss, 1991). It is no surprise that companies obtaining the support of VCs often experience sales growth, strategic alliance formation, initial public offerings, and acquisitions (e.g., Arikan & Capron; Gulati &

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Higgins, 2003; Higgins & Gulati, 2003; Lee, Lee, & Pennings, 2001; Ozmel, Reuer, & Gulati, 2013; Pollock & Gulati, 2007; Ragozzino & Reuer, 2011).

The early academic work that has examined the effects of the relationship between VCs with new ventures has frequently considered the involvement of VCs as a discrete feature, separating VC-backed new ventures from others and then tracking their subsequent developments. However, increasing attention is being paid to the question of *what* aspects of VC involvement influence the outcomes of new ventures that have *already* partnered with venture capitalist firms. This is an important question because entrepreneurs often face challenging decisions when considering a VC partner. For example, entrepreneurs must determine which VCs hold the ideal characteristics to help meet the new ventures' objectives, when the appropriate timing of VC entry should be, and how much capital is required to advance the firm. These decisions can be extremely consequential for entrepreneurial ventures. For instance, Hsu (2004) shows that reputable VCs are able to acquire stakes in a company at a 10–14% discount compared to less reputable ones. Therefore, notwithstanding the several benefits that may result from the association with prominent VCs, entrepreneurs may stand to bear substantial costs for these partnerships, as well. In this respect, despite the strides made by recent work in this area (see for instance, Pollock, Chen, Jackson, & Hambrick, 2010; Reuer, Tong, & Wu, 2012), there remains substantial scope for work aimed at explaining the link between the properties of VCs and the outcomes experienced by entrepreneurial ventures.

In this paper, we aim to build on this recent literature. We draw from signaling theory and depart from solely considering the presence of a VC in a new venture as a dichotomous signal of the venture quality. Rather, we examine several aspects of the relationship between VCs and new ventures in terms of the signaling properties that might explain two key entrepreneurial outcomes: acquisitions and initial public offerings (Dushnitsky & Shaver, 2009; Gompers & Lerner, 2004; Guler, 2007). More precisely, we examine five signals: the prominence of the VCs affiliated with a new venture, the number of VCs, the timing of their entries, the duration of their investments, and their total disbursements into new ventures. We hypothesize that more prominent VCs, a higher number of VCs, earlier VC involvement, shorter investment duration, and larger total disbursements will be associated with a greater probability of entrepreneurial exit via an acquisition or an IPO. Our investigation draws from a sample spanning 26 years (i.e., 1985–2010) containing nearly 3,600 VC-backed entrepreneurial companies. Overall, the results provide strong support for our predictions. Additional analyses also reveal that the five signals we explore play out differently depending upon whether the outcome considered is an acquisition or an IPO. Namely, while VC prominence and the total amount disbursed by VCs appear to be strong predictors of initial public offerings, they do not show the same effects on acquisitions. Moreover, the number of VCs involved in a new venture significantly explains the acquisition exit outcome, but not the exit by initial public offering. These findings and several others are discussed in detail later in the paper.

This paper represents a contribution to the entrepreneurship literature for at least two reasons. First, we expand on recent work that has studied the effects of the micro features of institutional affiliations on entrepreneurial companies' development (Ozmel et al., 2013; Pollock et al., 2010; Puri & Zarutskie, 2012; Reuer et al., 2012). Our analysis disaggregates the signals embedded in VC affiliation with greater precision than before, allowing us to forge a clearer link between the involvement of venture capitalist firms and the outcomes experienced by new ventures. From a practical standpoint, the results in this paper also allow us to offer prescriptive guidance to start-ups contemplating partnering with VCs early after founding. Second, our examination of two separate outcomes—acquisitions and IPOs—provides evidence of how the features of VC affiliation may result in different upshots for new ventures.

Theory and Hypothesis Development

Information Asymmetry and Signaling

The importance of asymmetric information in business has been discussed extensively by financial economists. For example, Akerlof (1970) used the market for used cars as the context to illustrate the implications of the adverse selection problem in product markets. He showed that unless appropriate—and costly—remedies are put in place, the market may not clear, or buyers may face the risk of overpaying for a used vehicle. As another example, Spence (1973) argued that prospective employees might be able to deploy their educational achievements as tools (i.e., signals) for companies to distinguish promising hires from others. Accordingly, information signals allow better quality employees to earn higher wages *vis-à-vis* lower quality ones. Since the work of Akerlof and Spence, the usefulness of information signals as remedies to the adverse selection problem has been studied in large number of settings, and Bergh, Connelly, Ketchen, and Shannon (2014) provide a useful review and a discussion of this vast literature.

To the extent that the benefits of signals are greatest when direct information on an item cannot be obtained, the hazards of adverse selection will be especially severe when applied to the context of new ventures—for several reasons: New ventures face challenges stemming from information asymmetry (Gompers & Lerner, 2004; Graebner, 2009), owing to the liabilities of newness, which raise search and valuation costs for investors seeking out attractive opportunities (Hagedoorn & Sadowski, 1999; Hopp & Lukas, 2014; Mitchell & Singh, 1992; Mody, 1993). Also, since the value of new companies is tied predominantly to growth expectations and the figure of the entrepreneur—rather than assets in place—the risks of due diligence tend to be higher, as well (Barzel, 1987; Shane & Cable, 2002; Shane & Stuart, 2002). Lastly, knowing the true intent of new ventures' managers may be challenging (Elitzir & Gavios, 2003), as information asymmetries provide an incentive for entrepreneurs to exaggerate their prospects and misrepresent the value of their companies (Ravenscraft & Scherer, 1987). Therefore, outside investors often look for signals before making investments in order to reduce the problem of adverse selection (e.g., Certo, 2003; Higgins & Gulati, 2006; Lester, Certo, Dalton, Dalton, & Cannella, 2006; Zimmerman, 2008).

In the same spirit as Spence's seminal piece, we use the characteristics of the involvement of venture capitalist firms in entrepreneurial companies as signals of quality and predictors of acquisitions and initial public offerings. Specifically, we use the prominence of the VCs, the number of VCs invested in a company, the timing of their first disbursement, the total amount disbursed, and the duration of their investment as signals of the properties of entrepreneurial companies. These signals are both observable and costly to obtain, as posited by the theory (Connelly, Certo, Ireland, & Reutzel, 2011). It is well known that new ventures bear significant costs when they partner with VCs (Guler, 2007; Hsu, 2004) and this, in turn, increases the effectiveness of the signal inherent in the VC affiliation, since investors are not easily fooled by cheap signals (Lee, 2001). Below we develop individual hypotheses with respect to each signal and the likelihood of the acquisition of the entrepreneurial venture, or the entrepreneurial venture's IPO.

IPOs and Acquisitions as Exit Events

Before we turn to our predictions, there are a few important points to be made with respect to the two exit outcomes we investigate. First, although not without exceptions (see Manigart & Wright, 2013, for a discussion), IPOs and acquisitions are generally viewed as attractive events for entrepreneurial ventures. For example, IPOs allow firms to improve their liquidity (Haveman & Khaire, 2004), gain new managerial skills via the introduction of new teams (Wasserman, 2003), and scale up their product offerings (Aldrich, 1999). For their part, acquisitions pave the way for the creation of synergies, access to new markets, and the realization of scale economies that would likely be unavailable to the new venture alone (Larsson & Finkelstein, 1999; Seth, 1990). However, it is plausible that in some cases, these events may be undesirable for entrepreneurs.

For example, Puri and Zarutskie (2012) argue that distressed sales, buy-backs, liquidations, and bankruptcies are one of the least understood aspects of exit, and in their paper, they test whether VCs might engage in fire-sale acquisitions to liquidate bad investments. However, they do not find support for their fire-sale hypothesis. Furthermore, Masulis and Nahata (2011) report that when a VC backs a new venture in its seed or early life stages, acquisitions of these firms result in higher takeover premiums for targets. Although the assumption of exit being successful is not central in our work, we should note that we only focus on *full* acquisitions, which require both VCs and entrepreneurs to sell their equity in the venture. Consequently, we do not count the sale of partial ownership by VCs as exit events in our data, as we will thoroughly discuss later in the paper.

A second important point is that although both represent exit vehicles for entrepreneurs, acquisitions and IPOs are also fundamentally very different corporate events. For example, while the former are one-shot transactions involving individual buyers with certain strategic objectives to be extracted from a deal, IPOs are capital-raising endeavors appealing to the broader investment community and they are often followed by subsequent secondary offerings. As a second example, the processes leading up to either outcome are different, as well, with acquisitions often featuring private negotiations between the parties, whereas IPOs entail road shows and an elaborate information disclosure process. Based on these and other differences, there is a tension with respect to whether it would be appropriate to aggregate these two outcomes conceptually, or otherwise develop separate predictions with respect to the likelihood of exit by an acquisition or an IPO. We opted for the latter solution, although the theoretical arguments developed to arrive at the predictions are the same, because acquisitions and IPOs are the two main exit routes for entrepreneurs and VCs alike (Manigart & Wright, 2013).

VC Prominence as a Signal

The association with VCs can alleviate the problem of asymmetric information facing prospective investors by offering a screening mechanism for investors interested in a new venture. Moreover, the most prominent VCs will be the ones selecting the best new ventures to fund, as venture capital firms must build a credible reputation, in order to ensure their ability to engage in repeat business over time (Gompers, 1996; Nahata, 2008). Those firms that have been successful at delivering the intended objectives to entrepreneurs will be the ones most sought after by new ventures in future deals. Therefore, on one hand, the most prominent VCs will be the ones with “first dibs” into the most promising entrepreneurial prospects, but there will be reciprocity in this relationship, as promising new ventures will also seek out the VCs with the most prominence. Hence, we expect that the prominence of the venture capital and the (unobserved) quality of the new venture will be highly correlated.

There is a growing body of work that has underscored the importance for entrepreneurs to partner with credible venture capitalists (Lee, Pollock, & Jin, 2011; Reuer et al., 2012). For example, Manigart, Baeyens, and Van Hyfte (2002) study the survival rates of a sample of 565 Belgian VC-backed companies and find that being backed by experienced and reputed venture capitalists matters more than receiving VC capital in the first place. As another illustration, Chang (2004) finds that when new ventures partner with VCs featuring a track record of taking entrepreneurial companies public, the time to IPO of these new ventures decreases significantly. In turn, after controlling for endogeneity problems, Sørensen (2007) brings corroborating evidence of how experienced venture capitalists provide a sorting mechanism such that the companies they fund are more likely to go public and Nahata (2008) shows that reputable VCs lead to quicker and more successful exit. Recent work has also provided evidence of several other positive performance effects tied to VC prominence (see for instance Gulati & Higgins, 2003; Hsu, 2006; Krishnan, Ivanov, Masulis, & Singh, 2011; Reuer et al.).

Although the affiliation with highly prominent venture capitalists can be rewarding, these partnerships can also be costly to obtain, since prominent VCs hold substantial bargaining power *vis-à-vis* entrepreneurial companies. In fact, research has shown that they tend to obtain a discounted value on their investments ranging from 10% to 14% of the value of the company (Hsu, 2004). Paralleling the logic used by Akerlof (1970), who discussed the use of costly remedies by good-quality sellers, such as warranties and money-back guarantees as signals to reduce information asymmetry, the best entrepreneurial companies will be willing to bear the costs of securing the involvement of the most prominent VCs, in order to offer a stronger signal of their value to prospective investors. In contrast, lower quality new ventures—the ones most likely to leverage on information asymmetry and misrepresent their prospects—will be unable to replicate this strategy (Akerlof). Therefore:

Hypothesis 1a: The higher the prominence of the VC, the greater the likelihood that the new venture will be acquired after its founding.

Hypothesis 1b: The higher the prominence of the VC, the greater the likelihood that the new venture will go public after its founding.

Number of VCs Invested in the New Venture

Our second hypothesis positively relates the likelihood of exit to the number of VCs invested in a new venture. The presence of a VC in a new venture has been found to be an unambiguous indicator of the qualities of the latter (Gompers, 1996)—and prior work has shown that (1) venture capitalists are very selective of the investments they take on (Megginson & Weiss, 1991) and (2) VC-backed firms experience superior outcomes *vis-à-vis* their counterparts, *ceteris paribus* (Higgins & Gulati, 2003; Hsu, 2006; Krishnan, Chemmanur, & Nandy, 2011; Lee et al., 2001; Pollock & Gulati, 2007). Thus, when a VC supports a new venture, the information signal embedded in this action should be perceived favorably.

The logic above indicates that the strength of the information signal issued through the receipt of VC funds will grow positively with the number of VCs funding the new venture. The only instance in which this argument would not hold is if all venture capital firms were price takers endowed with identical resources, capabilities, and information. Under these circumstances and conditional on at least one VC funding a new venture, the marginal signaling value of the next VC funding the venture would be zero, as it would offer no additional information to outsiders. However, the observable heterogeneity characterizing the VC industry does not support this scenario and even if the information benefit of additional VCs may be decreasing with every new firm's funding the entrepreneurial venture, we predict that as the number of VCs invested in a firm increases, the strength of the signal inherent in the VC endorsement will also increase, hence:

Hypothesis 2a: The larger the number of VCs invested in the firm, the greater the likelihood that the new venture will be acquired after its founding.
Hypothesis 2b: The larger the number of VCs invested in the firm, the greater the likelihood that the new venture will go public after its founding.

Timing of VC Investments

The next feature of the VC–new venture relationship that we investigate is the timing of the initial capital injection made by a VC into the new venture. When new ventures are founded, they face a great deal of uncertainty about their future (Gompers & Lerner, 2004). As a result, these ventures find it hard to procure capital and resources from prospective investors and other stakeholders, since these parties face adverse selection problems *vis-à-vis* the new venture. Consequently, it is imperative for a new venture to produce information that will help the investment community assuage concerns stemming from asymmetric information (e.g., Arrow, 1974). One signal that may relay positive information and in turn reduce problems of information asymmetry could be a new venture's ability to promptly establish an affiliation with a VC. Venture capitalists become directly involved in new ventures not only by providing funds, but also by taking active roles in decision making, contributing their network of relationships, hiring personnel, etc. (Brav & Gompers, 1997)—and providing this support early on should positively impact the entrepreneurial venture's prospects. Given the high level of uncertainty surrounding new ventures, and based on the expertise and selectivity of VCs, an early commitment made by a VC to a new venture sends a valuable signal. In other words, while all new ventures face the challenges inherent in the liability of newness (Freeman, Carroll, & Hannan, 1983), this problem becomes less and less relevant as the new venture builds its reputation and track record in the marketplace. In contrast, the younger the new venture, the more valuable the signal offered by its affiliation with a VC, because the information asymmetry problem tends to be most severe at the onset of the entrepreneurial cycle (McGrath, 1999). Therefore:

Hypothesis 3a: The earlier the new venture receives VC funding, the greater the likelihood that the new venture will be acquired after its founding.
Hypothesis 3b: The earlier the new venture receives VC funding, the greater the likelihood that the new venture will go public after its founding.

Duration of VC Investments

We now turn to the duration of the VC investment in a new venture. There are at least three reasons why investment duration should provide a useful signal for prospective investors in the new venture. First, an early stage new venture presents significant monitoring costs, since there tend to be fewer checks and balances within the venture—and the equity position of entrepreneurs is junior to the equity position of VCs, which confers to entrepreneurs what is tantamount to a long position in a call option (Gompers & Lerner, 2004; McGrath, 1999). Because of this, when facing incentive misalignment problems, VCs often spread their investments over time, because staged capital infusions affords VCs a control mechanism and extending the duration of funding helps address the general need to monitor entrepreneurs (Sahlman, 1990). Accordingly, when the presence of VCs in new ventures stretches over extended amounts of time, markets may interpret this signal to mean that VCs are facing challenges getting the entrepreneur to act in the best interest of the venture. In contrast, shorter investment durations would likely signal to prospective investors that the new venture has appropriate internal governance. Given that the agency problem is one of the most prominent hazards of investing in entrepreneurial companies (i.e., Gompers & Lerner), information signals that alleviate this concern to prospective investors in merger and acquisition (M&A) and IPO markets can be valuable.

A second reason why the duration of the VC investment in a new venture might offer an information signal on the value of the latter is that as a new venture ages, the novelty of its business idea is likely to decrease over time. Meanwhile, the likelihood that new competitors will emerge and threaten the uniqueness of the new venture will also increase (Carow, Heron, & Saxton, 2004). Given that the objective of VCs is to turn over entrepreneurial investments in as timely a manner as possible, prolonged participation in new ventures may be signals of their waning momentum. Finally, the information content of extended investment durations can be further exacerbated by the notion that some VCs may have a tendency to escalate their commitment to losing ventures (Guler, 2007). Therefore, aware of this point and of VCs' general business model, prospective investors may be induced to discount the value of new ventures with protracted affiliations with venture capitalist firms. Combined, the aforementioned points build the foundation for our next hypothesis:

Hypothesis 4a: The longer the duration of the affiliation of a new venture with VCs, the smaller the likelihood that the new venture will be acquired after its founding.

Hypothesis 4b: The longer the duration of the affiliation of a new venture with VCs, the smaller the likelihood that the new venture will go public after its founding.

Amount of Capital Raised

The amount of venture capital received by a new venture can also be an important information signal for prospective investors. Like all firms, VCs must make capital allocation decisions which inevitably come with opportunity costs. Namely, investing a larger portion of funds into a given venture will likely limit the extent to which a VC might be able to invest in other promising enterprises (Gompers & Lerner, 2004). Therefore, when VCs choose to invest significant amounts of funds into a new venture, they should do so either because they have determined that a particular investment corresponds to the best use of their resources, or because the scale of the investment is increasing and with it, the need to cover working capital requirements, investments in new infrastructures, etc. (Gompers & Lerner). Taking these points into consideration, along with the notion that VCs are very selective of which new ventures they invest in—it follows that the magnitude of the funds received by a new venture from VCs should be a positive indicator of the prospects of the new venture. More precisely, the larger the amount of funding received, the stronger (and the costlier) the signal created by the endorsement of venture capitalists in the new venture.

A second related consideration is that when VCs invest heavily in a new venture, the extent of their involvement tends to increase accordingly. In other words, while small investments may have VCs play a subsidiary role in the new venture, their level of commitment to the venture will increase proportionally with the magnitude of their investments (Guler, 2007). This implies that those new ventures that receive extensive funding from VCs will also benefit more from these firms' assistance with such activities as hiring of key personnel, marketing, operations, strategic planning, etc., than firms with smaller endorsements. In addition, VCs will be more likely to put into place control mechanisms that will induce the entrepreneur to work in the best interest of the new venture and bring the potential for agency conflicts to a minimum (Kaplan & Strömberg, 2003). Finally, the potential incentive held by VCs to "window-dress" a company for the purpose of maximizing its likelihood of exit will be reduced when there is greater investment in the venture, because the reputation capital of VCs increases with the extent of their participation in a new venture. Therefore, larger VC investments should work as stronger signals of the quality of the new venture to outside prospective investors. Hence, we predict the following:

Hypothesis 5a: The greater the magnitude of the VC funding, the greater the likelihood that the new venture will be acquired after its founding.
Hypothesis 5b: The greater the magnitude of the VC funding, the greater the likelihood that the new venture will go public after its founding.

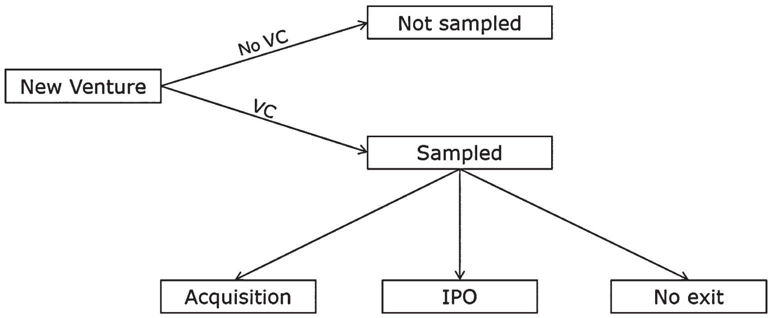
Methods

Sample

The main portion of the data was drawn from the Security Data Corporation database (SDC), which is a standard data source for research in the areas of entrepreneurship and corporate strategy. We used the Venture Xpert module of SDC to collect data on new ventures founded in the United States between 1985 and 2002 and operating in the manufacturing and service industries (i.e., SIC codes 20–39 and 70–89). We gathered further information on the VC investments made in these new ventures using the following criteria: First, given our focus on the relationship between new ventures and VCs, we only included investments made by VC firms. Thus, such entities as government affiliated programs, university programs, bank affiliates, pension funds, insurance firms, etc. were excluded from our sample. Second, we only considered companies that received at least one million dollars worth of total VC funding during the 10 years following their inception or at least until they experienced one of the two exit outcomes, as explained below. Third, we excluded companies whose birth was the result of a spinoff or a divestiture from an already existing company, as well as companies that did not experience exit but went on to pursue their own acquisitions as buyers, instead. This exclusion was motivated by the assumption that entrepreneurs seeking an IPO or an acquisition would likely pursue the support of VCs (because the latter also hold these objectives), while others would not. The inclusion of new ventures that became M&A buyers (a mere nine firms in our entire sample) did not alter the interpretation of the results, and later in the document, we provide additional details on this point. Lastly, following Puri and Zarutskie (2012), we only

Figure 1

Sample Description



included new ventures which SDC coded as being in "Early Stage," "Startup/Seed," "Expansion," and "Later Stage."

Figure 1 illustrates the broad criteria we followed when building our sample. It is worth noting that our choice to limit our sample to ventures that received VC support early in their lives does not create the potential for a selection bias problem, for several reasons. First, our investigation only surrounds VC-backed new ventures, and we do not extend our logic to ventures that have not received funding by venture capital firms. Second, prior work has needed to control for possible unobserved characteristics that may not only explain a key outcome variable experienced by new ventures, such as an IPO or an acquisition, but also the choice by VCs to invest in these ventures in the first place (see for instance Reuer & Ragozzino, 2012).

Unlike studies that examined whether the presence or absence of a VC led to performance differences for new ventures (see also Rosenbusch, Brinckmann, & Müller, 2013), our research focuses on the characteristics of the VC involvement, *conditional* on the VC presence. Therefore, we build on prior findings that have already shown that VC backing results in differential exit outcomes for new ventures and we try to determine whether specific aspects of the VC presence might themselves explain these outcomes differently. Finally, all of our theoretical variables are also conditional on the presence of a VC in the new venture and this makes the estimation of an eventual two-stage model correcting for endogeneity impractical. For example, when examining the prominence of a VC, it would be problematic to predict the likelihood of being backed by a VC in a first-stage model and then use the self-selection correction term in a model relating prominence of VCs within VC-backed firms to the likelihood of an acquisition or an IPO. Similar arguments apply to the number of VCs, as well as to the timing and magnitude of their investments.

Once we arrived at a baseline sample of new ventures, we tracked whether they became acquisition targets or whether they underwent an IPO within a decade of their founding. The data for this portion of the sample were drawn from the M&A and IPO modules of SDC and we limited our search to full acquisitions of the entrepreneurial ventures (i.e., 100% of the equity) and IPOs, thereby excluding all minority purchases, buybacks, privatizations, spinoffs, and equity carve-outs, as well as secondary offerings. Given that we followed the companies in our sample for 10 years after their inception, the complete data span the years 1985–2012 and after accounting for missing data, it comprises a total of 3,566 unique new ventures, of which 1,260 were acquired and 396 went through an IPO.

Model Specification and Measures

The basic structure of the multivariate statistical models is as follows:

$$\begin{aligned}\text{Entrepreneurial exit} = {} & \beta_0 + \beta_1 \text{ VC prominence} + \beta_2 \text{ Number of VCs investing} \\ & + \beta_3 \text{ Time to first VC investment} + \beta_4 \text{ Total amount invested} \\ & + \beta_5 \text{ VC duration} + \boldsymbol{\gamma} \text{ Controls} + \varepsilon \end{aligned} \tag{1}$$

Since we consider two forms of entrepreneurial exit, we constructed as many dependent variables, which we labeled *Acquisition exit* and *IPO exit.* Acquisition exit takes a value of one if the new venture was acquired within the first 10 years of its life and zero otherwise. Likewise, IPO exit takes a value of one if the venture experienced an IPO during the same time window, and zero otherwise. We estimated the models using logistic regressions, based on the dichotomous nature of the dependent variable. As a supplementary analysis, we also used partial likelihood using proportional hazards (Cox, 1972), which offers the benefits of accounting for timing information as well as for right-censored observations (Greene, 1997). This second set of models offered results that were extremely close to the logistic regression results and therefore we chose not to report them in detail below.

Explanatory Variables

VC Prominence. The first theoretical variable in our model is *VC prominence*. Our goal was to be as exhaustive as possible in the operationalization of this measure. As a consequence, we computed it in several ways. The first method, which is similar to one previously used in the literature (the commercial prominence measure developed by Gulati & Higgins, 2003; Stuart et al., 1999) involved three steps: First, we tracked the activity of the VCs contained in the Venture Xpert module of SDC, counting the number of companies in which a VC had invested in 5-year intervals (i.e., for any given year, we counted for the 5 years prior). Second, we identified the lead VC in each company as the VC that invested the greatest amount of dollars in the company from the time of the first disbursement to the last, as reported by SDC. Finally, using the year of the last disbursement as a matching criterion, we compared the total number of companies for the lead VC against the overall distribution for all venture capital firms and defined VC prominence as follows: We coded the variable as one if the number of companies was above the 75th percentile of the distribution, and zero otherwise. In order to gauge how the prominence of the VC affected the likelihood of exit, we also experimented with other cut-offs, such as the 50th and the 90th percentile, and we discuss the results of these additional steps later in the paper.

The second method by which we computed VC prominence was by replicating the steps outlined above, but rather than comparing the number of prior deals entered into by a VC, we used the value of their investments in all companies in the 5 years preceding the first investment in the focal new venture. Thus, this alternate measure places emphasis on the overall capital outlays implemented by VCs, rather than the number of investments. Third, we measured VC prominence by tracking the number of firms that the VC had led to either an acquisition or initial public offering in the 5 years preceding the first investment in the focal new venture (see for instance, Chang, 2004). This last measure of VC prominence is therefore tied to the ability by a VC to execute on its exit business plan. Clearly, each of the measures above presents distinct trade-offs, and later in the paper, we will present results using all of these specifications.

Number of VCs Invested in the New Venture. The second theoretical variable in the model is the number of different VCs that funded the new venture during the 10 years comprising our observation period. The data required to compute this variable was drawn from the information provided in Venture Xpert.

Timing and Magnitude Effects of VC Investments. In addition to measuring the prominence of the VCs involved in new ventures, we are interested in exploring whether the timing and magnitude of VC investments can also explain exit. First, we implemented a variable labeled *Time to first VC investment* that measures the number of years, since inception, that it took the new venture to secure VC funding. As discussed in the hypothesis section of the paper, we expect that more timely investments by VCs will send a stronger signal about the promise and quality of the underlying venture.

The next theoretical variable in the model is *VC duration*. Paralleling the logic used to motivate the inclusion of the time to first VC investment variable, we expect that the most attractive companies will be seen through by venture capitalists more quickly than investments made in new ventures with less promise. VC duration was computed as the number of years that elapsed between the time of the first VC investment in the new venture until the time of the last, and it is bounded between zero and 10 years.

The last theoretical variable in the model is *Total amount invested*, which is defined as the total dollar amount received by a company from all venture capitalists from the time of the first to the time of the last investment. The size of the aggregate investment into a new venture can also represent a useful signal for its prospects, because higher quality new ventures will likely receive more capital than lower quality ones. Although our data only comprises two broad industry segments (i.e., manufacturing and services), it is plausible that some industry segments defined at more fine-grained levels may hold capital requirements that are systematically higher than others. Therefore, besides introducing an appropriate industry control in the model, as discussed below, we also experimented with an industry-adjusted measure of total amount invested and found nearly identical results. The computation of the variables above was consistent with prior work (e.g., Gompers, 1996).

Control Variables

Besides the five theoretical variables, we included several controls that may also explain the dependent variables. The first two variables account for considerations tied to location and geography, which have been found to be highly relevant in entrepreneurial outcomes (Audretsch & Feldman, 1996; Gupta & Sapienza, 1992; Jaffe, Trajtenberg, & Henderson, 1993; Lerner, 1995; Sorenson & Stuart, 2001; Stuart & Sorenson, 2003). The variable *Company in California* is an indicator that takes a value of one if the entrepreneurial venture was situated in the state of California, and zero otherwise. Not only is California the recipient of the majority of VC funds in the United States (i.e., Gompers & Lerner, 2004), it is also and most importantly the state in which many acquisition targets and newly public companies are located. The second control is *Same state*. This variable takes a value of one if the lead VC and the new venture were located in the same state, and zero otherwise. Prior work has shown that VCs tend to prefer proximate investments over remote ones (Lerner, 1995), because they are better able to value and monitor local ventures. Therefore, if a VC decides to invest in a distant venture, this should represent a stronger signal about the quality of the venture holding all else equal, as recent work has also shown (Ragozzino & Reuer, 2011).

The next control is a count measure of all the equity (i.e., equity swaps, joint ventures) and nonequity alliances in which the new venture had engaged during the 10-year observation period. If the company experienced exit prior to the 10 years, we counted until the year of the exit. The existence of prior alliances can reduce the effects of asymmetric information with which investors are faced when evaluating a prospect (Hagedoorn & Sadowski, 1999; Mitchell & Singh, 1992; Mody, 1993; Porrini, 2004), and they can help to expand a company's network of exchange partners, thereby enhancing its credibility (Stuart et al., 1999) and increasing its likelihood of being acquired or going public. The data for the computation of this variable were drawn from the SDC database.

Finally, we wanted to account for industry- and time-specific effects that may also explain the exit outcome experienced by entrepreneurial ventures. First, we included a control *Company manufacturing*, which allowed us to separate the two broad categories of firms comprising our sample (i.e., manufacturing and services) and determine whether either sector results in a different probability of acquisition or IPO for the focal company. We also experimented with more fine-grained industry segments and found that these steps did not alter the results significantly. Second, we implemented year fixed effects to account for the impact of broad macroeconomic determinants on the probability of exit by the entrepreneurial ventures. Specifically, we include one indicator variable for each of the 16 years in our sample, minus one (i.e., 1986–2000). Below we discuss the results of the analyses and then we check their robustness by introducing a host of supplementary tests and models.

Results

Table 1 presents descriptive statistics for the nonbinary variables in our models as well as a correlation matrix. Table 2 offers supplemental analyses that split the sample between companies that experienced exit from others, and then compares the mean values of the theoretical variables across these categories. Of the 3,566 observations in the dataset, 40% experienced exit through an acquisition and 17% went through an IPO within 10 years of founding. These figures are higher than the ones reported by Puri and Zarutskie (2012), who report an acquisition rate of 24% and an IPO rate of 7%. We suspect that our focus

Table 1

Descriptive Statistics and Correlation Matrix[a,b]

Variable	Mean	SD	1	2	3	4	5	6	7	8
1. VC prominence (count)	.70	.46								
2. VC prominence (value)	.22	.41	.33***							
3. VC M&A exit experience	15.05	15.05	.48***	.66***						
4. VC IPO exit experience	9.79	11.96	.42***	.59***	.69***					
5. Number of VCs	6.58	4.33	.36***	.27***	.25***	.19***				
6. Time to first VC investment	1.50	1.96	−.15***	−.15***	−.14***	−.10***	−.28***			
7. VC duration	3.52	2.53	.18***	.12***	.20***	−.02	.50***	−.34***		
8. Total amount invested[c]	35.24	44.04	.20***	.18***	.22***	.09***	.50***	−.17***	.32***	
9. Company alliances	.60	1.25	.09***	.07***	.04**	.07***	.16***	−.02	.13***	.08***

** $p < .01$, *** $p < .001$. [a] n = 3,566. [b] n = 3,170 and n = 2,306 for Acquisition exit and IPO exit, respectively. [c] In millions of dollars.

Table 2

Comparative Tests of Theoretical Variables by Exit Mode

	Acquisition			IPO		
Independent variables	No (N = 1,910)		Yes (1,260)	No (1,910)		Yes (396)
Lead VC prominence (count, 50th pctl)	.84 (.36)		.84 (.37)	.84 (.36)		.92 (.27)
χ^2		.15			15.65**	
Lead VC prominence (count, 75th pctl)	.68 (.46)		.70 (.46)	.68 (.46)		.82 (.38)
χ^2		.42			29.51***	
Lead VC prominence (count, 90th pctl)	.48 (.50)		.50 (.50)	.48 (.50)		.67 (.47)
χ^2		1.27			46.14***	
Lead VC prominence (value, 75th pctl)	.19 (.40)		.21 (.41)	.19 (.40)		.35 (.48)
χ^2		1.79			44.50***	
Lead VC M&A exit experience	14.60 (14.68)		14.81 (15.57)	14.60 (14.68)		18.05 (14.85)
t-test		−.38			−4.25***	
Lead VC IPO exit experience	8.98 (11.21)		8.55 (11.07)	8.98 (11.21)		17.63 (14.91)
t-test		1.05			−10.93***	
Number of VCs	6.48 (4.35)		6.21 (3.98)	6.48 (4.35)		8.27 (4.87)
t-test		1.77†			−6.79***	
Time to first VC investment	1.69 (2.18)		1.28 (1.58)	1.69 (2.18)		1.28 (1.82)
t-test		6.19***			3.97***	
VC duration	3.83 (2.62)		2.80 (2.16)	3.83 (2.62)		4.29 (2.64)
t-test		11.99***			−3.19**	
Total amount invested[a]	34.77 (44.68)		32.41 (40.38)	34.77 (44.68)		46.52 (50.02)
t-test		1.54			−4.33***	

† $p < .10$, ** $p < .01$, *** $p < .001$. [a] In millions of dollars. pctl, percentile.

on the manufacturing and service industries, along with the minimum U.S.$1 million investment requirement we set and the somewhat dissimilar timeframes we use explains the differences between our results and theirs. Also, we should note that in 127 cases, the new ventures in our sample went public and then became acquisition targets, which explains the positive and significant correlation between the two outcome variables (i.e., $p < .001$). We coded these transactions as IPOs, because IPOs offer numerous signaling opportunities of their own (Pollock & Gulati, 2007), which may make the effects of VC-related signals less relevant.

About 70% of the firms in our sample were led by a prominent VC by prior deal counts, although this number drops to only 22% when we use capital investments as a proxy for VC prominence. Moreover, the average lead VC had experienced 15 acquisitions and about 10 IPOs in the 5 years preceding their first investment in the focal firm, and the mean number of VCs invested in a firm at the time of the last investment was 6.6. It is worth noting that the various measures of VC prominence we adopt are all positively and significantly correlated with each other (all $p < .001$) and that the correlations of the VC prominence proxies with the two exit types are very different, seemingly being much more relevant for IPOs than for acquisitions. Table 2 offers additional insights on these relationships. The chi-square and *t*-tests shown indicate a marked difference between firms that experienced an initial public offering and those that did not with respect to all measures of VC prominence (i.e., all *p*-values between $p < .01$ and $p < .001$). In contrast, none of the VC prominence proxies appears to be different when we dichotomize the sample between firms that experienced no exit and firms that were acquired in M&A markets (i.e., all n.s.).

Turning our attention to the other theoretical variables, the Time to first VC investment variable is highly correlated with both the acquisition and the IPO exit indicators, which brings preliminary—albeit merely descriptive—support for our predictions (i.e., both $p < .001$). The *t*-tests shown in Table 2 further corroborate these descriptive results (i.e., both $p < .001$). The VC duration and the Total amount invested variables present somewhat interesting results. For instance, firms that experienced an IPO have statistically longer relationships with VCs (4.3 versus 3.8 years, $p < .01$) and VCs invest substantially more in these firms, too (i.e., U.S.$46.5 million versus U.S.$34.8 million, $p < .001$). In contrast, the duration of VC investments is substantially shorter for firms exiting via acquisitions than for firms experiencing no exit (i.e., 2.8 versus 3.8 years, respectively, $p < .001$), although the magnitude of VC investments in these firms does not differ statistically between the two classes of firms (i.e., U.S.$32.4 million versus U.S.$34.77 million, n.s.). We will explore these relationships in greater detail in the ensuing multivariate analysis.

With respect to the VC duration variable, we should point out the following: One might assume that if the new venture exits, then by definition the VC investment periods will be shorter, because the VC–new venture relationship will (by definition) stop. In turn, this may raise the concern that essentially the dependent variable (DV) (i.e., exit) and the independent variable (IV) (i.e., VC duration) measure the same thing. When we compared the distributions of VC duration based upon whether an acquisition or an IPO outcome was experienced, we found that they were relatively similar. For instance, we found that by the third, sixth, and ninth years from the founding of the new venture, roughly 66%, 94%, and 100% of the firms that were bought out had already received their last VC disbursement and these numbers are 49%, 81%, and 97% for their counterparts. Perhaps the similarities between firms that went public and firms that stayed private during the observation period are even more surprising: Namely, the third-, sixth-, and ninth-year percentages of firms receiving their last VC disbursement before their IPOs were 44%, 76%, and 97%, as opposed to 50%, 82%, and 97% for firms that did not go public. Formal Kolmogorov–Smirnov tests did not allow us to reject the null of equal distributions for either comparison (i.e., $p < .001$ and $p < .01$ for acquisitions and IPOs, respectively). However, the numbers above should alleviate the concern that our 10-year window leaves out many firms that would continue to receive VC support past the tenth year and eventually experience exit, because it appears that the greatest majority of VC-backed firms stop being funded well within the time window we chose.

The correlations among the covariates are also noteworthy. First, it seems that when VCs invest in a company early after its founding, they tend to invest less and for less time (i.e., both $p < .001$). Similarly, longer durations are positively related to greater dollar investments (i.e., $p < .001$). There are also significant positive correlations between the prominence of a VC and its inclination to invest in new ventures situated in California or in its same state (i.e., $p < .001$ and $p < .01$, respectively). Interestingly, when investing in the same state, VCs tend to become involved in a new venture earlier in its life, disburse less money, and invest for longer periods of time than in cross-state transactions (i.e., all $p < .001$). These findings are consistent with arguments of asymmetric information that suggest that proximity allows for better screening of potential opportunities and of monitoring of existing investments by VCs (Coval & Moskowitz, 1999, 2001).

Table 3 presents the results for the acquisition exit models, while Table 4 estimates the same models for the exit by IPO. We should note that when estimating the probability of

Table 3

Logit Models—VC Characteristics and Exit by Acquisition[a,b]

Acquisition exit mode (n = 1,260)	I	II	III	IV
Intercept	−.01 (.10)	.48 (.45)	.47 (.45)	.45 (.45)
Lead VC prominence (count, 75th pctl)		.11 (.09)		
Lead VC prominence (value, 75th pctl)			.09 (.10)	
Lead VC M&A exit experience				.00 (.00)
Number of VCs		.03* (.01)	.03* (.01)	.03* (.01)
Time to first VC investment		−.14*** (.02)	−.14*** (.02)	−.15*** (.02)
VC duration		−.27*** (.02)	−.27*** (.02)	−.27*** (.02)
Total amount invested[b]		.04 (.05)	.05 (.05)	.05 (.05)
Company in California	−.05 (.08)	−.12 (.09)	−.12 (.09)	−.11 (.09)
Same state	.06 (.08)	.06 (.09)	.06 (.09)	.06 (.09)
Company alliances	.03 (.03)	.07* (.03)	.07* (.03)	.07* (.03)
Company manufacturing	−.11 (.08)	−.06 (.09)	−.06 (.09)	−.06 (.08)
Time fixed effects	186.24***	139.71***	139.24***	138.31***
Model χ^2	214.09***	428.46***	427.94***	427.20***

* $p < .05$, *** $p < .001$, one-sided tests. [a] n = 3,170. [b] This variable was logged for the model estimation to remedy skewness.
pctl, percentile.

Table 4

Logit Models—VC Characteristics and Exit by IPO[a]

IPO exit mode (n = 396)	I	II	III	IV
Intercept	−3.52*** (.26)	−8.82*** (.82)	−8.43*** (.81)	−8.05*** (.81)
Lead VC prominence (count, 75th pctl)		.35* (.16)		
Lead VC prominence (value, 75th pctl)			.41** (.14)	
Lead VC IPO exit experience				.03*** (.00)
Number of VCs		.01 (.02)	.01 (.02)	.01 (.02)
Time to first VC investment		−.17*** (.04)	−.16*** (.04)	−.15*** (.04)
VC duration		−.10*** (.03)	−.10** (.03)	−.08** (.03)
Total amount invested[b]		.55*** (.08)	.53*** (.08)	.49*** (.08)
Company in California	.19 (.13)	−.06 (.14)	−.08 (.13)	−.19 (.14)
Same state	−.03 (.14)	.05 (.13)	.04 (.14)	.02 (.13)
Company alliances	.22*** (.04)	.17*** (.04)	.17*** (.04)	.16*** (.03)
Company manufacturing	.75*** (.12)	.61*** (.13)	.61*** (.13)	.67*** (.13)
Time fixed effects	103.14***	153.72***	150.29***	113.81***
Model χ^2	229.60***	342.96***	346.53***	383.41***

* $p < .05$, ** $p < .01$, *** $p < .001$, one-sided tests. [a] n = 2,306. [b] This variable was logged for the model estimation to remedy skewness.
pctl, percentile.

one exit type, all instances of the other exit type are excluded from the sample and this explains the different sample sizes for the acquisition and the IPO models (i.e., 3,170 and 2,306, respectively). The first column of each table reports the results from controls-only specification, while columns II–IV report the results of the full models using different constructs for VC prominence. All models are highly significant on an overall basis (i.e., all $p < .001$), and in all cases, the introduction of the theoretical variables yields a statistically significant improvement over the baseline specification (i.e., all $p < .001$).

The first set of hypotheses predicted a positive relationship between VC prominence and the likelihood of acquisition and IPO exits. We find mixed support for these predictions. The results in columns II–IV in Table 3 do not provide significant indication of the role of VC prominence on acquisition exit (i.e., all n.s.). In contrast, we find evidence suggesting that VC prominence explains the probability of an initial public offering in the first 10 years in the life of a new venture (i.e., Table 4, column II). This result holds for all VC prominence specifications (i.e., p-values between $p < .05$ and $p < .001$) and it indicates that the odds of going public for a new venture funded by a prominent VC are roughly 42% greater. As a robustness check, we re-estimated the models replacing the VC prominence measure that used the 75th percentile count cut-off, with both a more conservative (i.e., 50th percentile) and a more aggressive (i.e., 90th percentile) specification. We also used the total prominence of all VCs involved in the new venture, as opposed to the prominence of the single lead VC, computed as the total number of prior investments entered into by all VCs invested in the focal firm. The results of these additional analyses do not change the general interpretation of the main finding, which is that VC prominence explains the acquisition exit very marginally—which runs counter to our prediction—and the IPO exit significantly, as predicted.

Turning to our second set of hypotheses, the Number of VCs variable, the acquisition model (i.e., Table 3, column II) shows evidence that the number of venture capital firms invested in a firm is associated with a higher likelihood of an acquisition of the latter (i.e., $p < .05$) and that the odds of an acquisition are about 3% greater for every additional VC that funds the new venture. In contrast, no such a relationship is revealed in Table 4, which predicts exit by IPO. Given the relatively high correlation coefficient existing between the Number of VCs and the Total amount invested variables, we wished to explore the possibility that acquisition markets use the signals of quantity of VC investors and quantity of funds disbursed by VCs interchangeably, while IPO markets may be more discerning of the actual funds invested in a venture, regardless of how many VC firms are involved. We estimated a model with an interaction term between the Total amount invested variable and the Number of VC firms variable and the results offered convincing support for the substitutability hypothesis in M&A market, as the two main theoretical variables retained their signs and significance, and the interaction term was negative and significant, as well (i.e., $p < .01$). We did not find similar evidence in the IPO exit model, however.

As an additional test, we estimated models without the Total amount invested variable, in order to gauge whether the IPO markets viewed the number of VCs and the total amount invested as redundant signals, as the high correlation coefficient might suggest. The results of these models indicated that the Number of VC firms was positive and highly significant (i.e., $p < .01$ and $p < .001$ for the acquisition and IPO exit models, respectively). Lastly, we wanted to determine whether the signaling value of VCs funding a new venture decreases with each additional venture capital firm entering the investment. In order to test this un-hypothesized idea, we introduced a squared-term in the models and found that in the case of exit by acquisition, it was indeed negative and significant (i.e., $p < .05$). Jointly with the direct effect retaining its positive sign and significance (i.e., $p < .01$), we therefore find evidence of diminishing returns in the Number of VCs variable. However, neither variable reached conventional statistical significance in the IPO exit models. Overall, the evidence at hand is in support of the hypothesized relationship with respect to the acquisition outcome and only indirectly supporting the hypothesis tying the Number of VCs variable to IPO exit.

Our third set of hypotheses predicted a negative relationship between the time it takes a new venture to receive initial funds from a VC and the likelihood of exit. We find support for this prediction. The models in Tables 3 and 4 indicate a highly significant coefficient for the Time to first VC investment variable (i.e., all coefficients $p < .001$). It is also interesting that, besides their significances, the magnitudes of the coefficients are also similar across the models in Tables 3 and 4, showing that for every year after founding that a company does not receive its first VC investment, the odds of exit decrease by about 13% and 15% for acquisitions and IPOs, respectively. Our fourth set of hypotheses predicted that longer VC investments would result in a lower likelihood of exit and the results in Tables 3 and 4 offer strong support for this prediction (i.e., $p < .001$). However, even though all the coefficients are highly significant, the odds of acquisition exit as a function of VC duration are much greater than the odds of IPO exit. Specifically, the results show that for every additional year of permanence of VCs in the new venture, the odds of an acquisition decrease by about 24%. In contrast, the same delay decreases the risk of an IPO by about 10%. Thus, although we find support for our hypothesis, we also note the different economic effects of VC duration on the dependent variables.

Our last set of hypotheses predicted that the total amount invested in a company by VCs will increase the likelihood of exit. As in the case of the Number of VCs variable, we only find mixed support for this prediction. The models predicting IPO exit show very strong evidence of the importance of the Total amount invested variable, as the parameter estimates are very significant across all specifications (i.e., all $p < .001$), and their large coefficients also speak of the economic importance of this aspect of the firm–VC relationship. More precisely, for every increase in VC funding by one million dollars, the odds of going public increase by roughly 60%. In contrast, while positive, the coefficients of the Total amount invested variable never reach statistical significance in the acquisition exit models, thereby failing to offer support for our hypothesis. We can therefore conclude that the total funds invested in a company by VCs only explain the likelihood of exit by IPOs.

Although many of our hypotheses were supported by the analyses, it is also apparent that the magnitude of the effects of the theoretical variables on exit are different, depending upon whether we examined acquisitions or IPOs. In order to determine whether the differences in the scale of the coefficients are attributable to random variation or whether they are systematic, we performed a competing risk analysis (i.e., Allison, 1995). This analysis consists of estimating a model that combines both exit types in a single dependent variable, along with separate models that censor one type, while examining the other. Then, a test computing the difference between the log-likelihood of the combined model with the summation of the log-likelihoods of the two separate models is run, with the difference being a chi-square value with 46 degrees of freedom. The result of these steps yielded strong evidence that acquisition and IPO exits should be estimated in separate models (i.e., $\chi^2 = 472.63$, $p < .001$).

Finally, the control variables in the model deserve some commentary. First, we do not find evidence that issues related to the geographic location of the new venture affect its likelihood of exit in any of the model estimations. Second, a company's prior exposure to strategic alliances greatly affects its probability of being acquired or going public within the first 10 years of its life (models between $p < .05$ and $p < .001$). As before, the effect of this variable on exit depends upon the type of exit being considered. In fact, for every additional alliance by the company, the odds of an acquisition rise by about 7%, while the odds of an IPO increase by roughly 19%. Aside from these differences in the magnitude of the coefficients, these findings are consistent with prior work that has emphasized the effects of a company's network of partners on its credibility and prospects (Hagedoorn & Sadowski, 1999; Mitchell & Singh, 1992; Mody, 1993; Porrini, 2004; Stuart et al., 1999).

Robustness Checks

In this section, our objective is to determine whether the previously reported results are subject to variations when we change our assumptions or otherwise introduce additional covariates in the models. First, we wished to account for the possibility of sample selection bias when testing the VC prominence hypotheses. Our concern was that the new ventures in our sample that were endorsed by prominent VCs might have possessed unobservable attributes that also made them more likely to experience IPO or acquisition exit. If so, it would be necessary to control for this selection bias, in order to minimize the possibility of biased parameter estimates. Following conventional practice, we set out to estimate a probit model predicting the probability that a new venture would be chosen by a prominent VC and then included the appropriate self-selection control in our main models. The specification of the first-stage model was somewhat problematic, because our firms are all privately held and therefore very little information on their qualities could be obtained.[1]

Despite the challenges above, we specified a model with a dichotomous dependent variable that assumed a value of 1 if the new venture was endorsed by a prominent venture, and zero otherwise. The covariates of this model were: indicators of the firm's geographic location by state with high activity levels (Baker & Gompers, 2003), the geographic distance between the firm and the VC, the equity and nonequity alliance experience held by the firm, the capital gains tax rate (i.e., Gompers & Lerner, 2004), and the total rounds of VC funding received in the industry of the firm in the year in which the firm received its first investment. We do not present a table summarizing the estimation of this probit model (available upon request), but we should note that all the covariates were highly significant (all $p < .001$ except for the alliance variable, which was $p < .05$) and that the model was also highly significant overall (i.e., $p < .0001$). More importantly, none of the second-stage model coefficients experienced a significant departure from the results discussed above. Precisely, in the case of exit by acquisition, the augmented models still provide support for hypothesis 2, hypothesis 3, and hypothesis 4 (i.e., $p < .05$, $p < .001$, and $p < .001$, respectively). Similarly, the models predicting an IPO exit bring support for hypothesis 1, hypothesis 3, hypothesis 4, and hypothesis 5 (i.e., $p < .05$ for hypothesis 1 and $p < .001$ for the other hypotheses). These results closely mirror the ones previously reported. Moreover, the self-selection corrector was not significant in the acquisition model (i.e., $p < .27$) and only marginally significant in the IPO model (i.e., $p < .10$).

As a second robustness check, we wanted to revisit the time window through which we track new ventures since their founding. In the paper, we chose to use a relatively long 10-year window, because it offered the dual advantage of a more conservative test as well as of more eventful observations in our data. However, prior work in entrepreneurship has set the threshold to define new ventures from as few as 6 years to as many as 12 years (Brush, 1995; Brush & Vanderwerf, 1992; Covin & Slevin, 1990; Shrader, 1996; Zahra, 1996; Zahra, Ireland, & Hitt, 2000). Thus, in order to gauge how sensitive our findings are to the definition of newness we adopted, we built separate samples which tracked firms for 6 and 8 years after their founding. The results of the logistic models (available from the

1. At a more conceptual level, the lack of directly observable characteristics of our firms is precisely why signaling is a paramount mechanism in entrepreneurial markets. Given that prospective investors (whether acquisition buyers or equity investors) cannot gather the information to help them tell apart attractive from unattractive prospects, they rely on signals—such as the prominence of the VC invested in the firm—as substitute information that can help them make an educated decision (Capron & Shen, 2007; Puranam, Powell, & Singh, 2006).

authors upon request) do not indicate deviations from the original findings. In sum, our findings appear to be robust to the choice of time window used to track new ventures once they are established.

Third, in order to assess the probability of one outcome simultaneously to the probability of the other, we first estimated a multinomial logistic regression in which the acquisition and the IPO exit outcomes were alternate events and the default comparison outcome was no exit. This model (not shown and available upon request) did not show any departures from the main findings. As an additional step, we also estimated a bivariate probit model, as a way to investigate the relationship between exit types as a function of the VC involvement characteristics. If acquisitions and IPOs are interdependent, then the error terms in the two choice models may be correlated and this relationship should be accounted for econometrically. The bivariate probit model exploits the correlations among the disturbances (Greene, 1997), such that if the estimated correlation between the disturbances is zero—i.e., if acquisitions and IPOs are not substitutive events—the likelihood function is equivalent to those obtained when estimating two binomial probit models independently. When the correlation is significantly different from zero, separate estimation of the two equations yields inefficient estimates (Greene). In this case, the correlation coefficient was far from significant, thusly not allowing us to reject the null of independence (i.e., $p = .68$).

Turning our attention to the individual theoretical variables in the models, we performed several new tests aimed at determining whether our results depend upon the specific proxies we used, or whether the adoption of alternate proxies also allows us to maintain the general interpretation of our findings. First, we wished to dig deeper in the definition of VC prominence. Although we have already discussed a few alternate measures of this variable in the preceding section, we perform a few additional tests below. First, we wanted to also account for the possibility that VC prominence might be industry specific. For example, some VCs might specialize in a certain industry and/or geographic area, while others might cast a wider net when scouting investment opportunities. Thus, a comparison of these two types of VCs might result in a skewed evaluation of their prominence and experience. In order to address this concern, we reconstructed our VC prominence measure in two alternate ways. The first is an industry adjusted variable that counts the number of investments by the lead VC (i.e., the one with the largest investment in the focal firm) in the industry of the focal firm (defined at the three-digit SIC code level) during the 5 years preceding the observation. The second measure follows a similar approach but it replaces industry with geography. More precisely, in lieu of the industry, we counted prior investments made by the lead VC in the U.S. state of the focal firm.

It is worth noting that predictably these two variables were positively and significantly correlated with the other VC prominence variables described previously in the paper (i.e., all $p < .001$), although the magnitude of the coefficients never reached a value greater than .60 in any pairwise correlation. In order to test the different effects the industry- and geography-specific measures of VC prominence might bring about on the likelihood of exit, we ran the models substituting the original variable with these alternate constructs. All of the other theoretical variables retained their signs and significance after these additional estimations. The VC prominence variable remained virtually unchanged across all specifications. Specifically, in the model predicting acquisition exit, the VC prominence variable kept a positive sign but never reached conventional statistical significance (i.e., $p < .81$ and $p < .52$ for the industry and the geography variables, respectively). Similarly, in the models predicting IPO exit, all the covariates other than the VC prominence remained unchanged *vis-à-vis* the results previously presented. However, although the VC prominence variable stayed positive, its significance dwindled (i.e., $p = .172$ and

$p = .459$, respectively). Lastly, we built industry- and geography-based VC prominence measures that replaced deal counts with total amounts invested and found that these alternates explained the IPO exit (i.e., $p < .05$ and $p < .10$, respectively), but not the acquisition exit, as in our original results. Overall, these supplementary analyses indicate that the role of VC prominence on new ventures' exit depends upon the manner in which this variable is operationalized.

Discussion

This paper examines the involvement of VCs in new ventures as a predictor of exit by an acquisition or an initial public offering. Rather than separating new ventures with VC support from those without, as prior research has tended to do, we only consider ventures which have already gained endorsements by venture capitalists. This allows us to focus parsimoniously on the key characteristics found in the relationship between VCs and new ventures. The results provide overall support for the predictions that the prominence of VCs; the number of VCs invested in a company; along with the timing, duration, and magnitude of their participation play important roles in the likelihood that entrepreneurial companies experience an acquisition or an IPO within the first 10 years of their founding. We also show the different ways in which these features play out depending upon the type of exit we consider. Thus, while securing VC support is an important milestone in the life of new ventures (Gulati & Higgins, 2003; Lee et al., 2001; Ozmel et al., 2013; Pollock & Gulati, 2007; Ragozzino & Reuer, 2007), it is equally important to explore the properties of VCs and the way their ties with entrepreneurial companies form and evolve over time.

The findings we report in this paper introduce a number of implications for entrepreneurial companies as well as venture capitalist firms. For the former, our results demonstrate that the choices surrounding fund raising in general and partnering with a VC in particular can be extremely consequential *vis-à-vis* entrepreneurial objectives. Common wisdom suggests that the involvement of a VC should represent a homogenous and unequivocal signal of the quality of the underlying venture. In turn, this signal has been found to reduce information asymmetry with the venture's stakeholders, facilitating the attainment of its corporate objectives (Brav & Gompers, 1997; Carter & Manaster, 1990; Stuart et al., 1999). Therefore, it may follow that entrepreneurs should wholeheartedly welcome the advent of venture capital. However, we show that entrepreneurs are better served to be more discerning of this decision. More precisely, it is paramount for entrepreneurs to consider such issues as the VC prominence and number of the VCs willing to invest in their companies, the timing of their entry, the duration of their intended participation, and the magnitude of their financial contributions. These items should be carefully weighed against the goals sought by an entrepreneur, as they play a significant role in explaining the company's subsequent corporate activity.

Turning to the perspective of venture capitalist firms, this paper provides clear evidence that the likelihood of successful investments hinges on several factors. First, it appears that VCs with greater involvement in entrepreneurial firms are better positioned to see their investments through the desirable M&A outcome. Thus, all else being equal, VCs should intensify their presence in new ventures, in order to develop intangible qualities that markets seem to reward. In addition, we show that early involvement and short-lived investments in entrepreneurial companies are positively related to a greater likelihood of acquisitions and IPOs for the latter. As a consequence, VCs should seek out investment opportunities early on and inject the required capital to make a timely push toward the intended goals, perhaps even when the risks inherent in this strategy are comparatively greater than a wait-and-see approach. In contrast, VCs would be ill-advised to delay their investments in new ventures and stagger investments over extended amounts of time, as this conservative approach results in a lower probability of exit.

We also feel that there is scope for further investigation of why the five signals we examine play out differently for acquisition and IPO events. For example, the fact that VC prominence does not affect the likelihood of the former but that it significantly affects the likelihood of the latter is theoretically intriguing. Although we do not have the tools needed to investigate the reasons for these differences empirically, we can offer a theoretical explanation that might be worthy of further research. It is possible that the well-known problem of hubris, or overconfidence held by management in acquiring firms (i.e., Roll, 1986) may cause these firms to proceed with the purchase of new ventures even in the absence of the VC prominence signal. In contrast, the aggregate—and therefore plausibly less biased—judgment of prospective IPO investors may require this signal to assuage the information asymmetry problem surrounding new ventures.

Another theoretical variable that explains the likelihood of acquisitions and IPOs differently is the number of VCs funding a new venture. Specifically, while this variable is positively related to the acquisition outcome, it is not significantly related to the IPO outcome. The compounded effect of the networks of partners held by multiple VCs might be the culprit of this difference, as it may allow for a more effective identification of firms interested in acquiring the new venture and in turn, for the prompt successful liquidation of the VCs' investments. These are clearly anecdotal explanations and future empirical studies aimed at testing these ideas would be helpful.

Although this paper makes a preliminary attempt at studying how the various aspects of the involvement of VCs in new ventures ultimately affect the latter, much remains to be done before a complete picture can be drawn on this broad topic. For example, we only explore acquisitions and IPOs, while it is easy to argue that a number of other outcomes could be attractive for entrepreneurs. For instance, companies may choose to remain private and pursue organic growth for decades after their founding. Our focus on two specific entrepreneurial exits limits the extent to which our findings can be generalized to all other outcomes experienced by newly founded companies. Similarly, despite our attempt to be comprehensive in breaking down the various aspects of the VC–new venture relationship, data restrictions and our own limitations might have caused us to leave out other relevant dimensions of this relationship. Alternatively, there might be a number of other affiliations—those with angel investors or banks, for instance—which may also be instrumental in defining the exit outcomes we explore. We hope that our paper begins to uncover the set of determinants of entrepreneurial success, but we also recognize that future research that builds on our work and extends its message further would be highly beneficial.

Additionally, our focus on disaggregating signals and studying them at a more micro level will hopefully pave the way for additional research that is inspired by our approach. As we move forward in our work, we believe that there is a need to depart from using coarse instruments and to strive to measure the determinants of the exit outcomes experienced by new ventures with greater precision. Our effort is just a small step in this direction, and there are several other ones that come to mind. For example, one could conceivably examine how the entrepreneur's own background, education, prior experiences, ties, etc. might offer credible signals to angel investors and private equity firms and induce them to fund the startup at its onset or see it through its growth and eventual exit later in its existence. The exploration of these questions through the use of precise instruments to understand the cause-and-effect links leading to entrepreneurial exits will be helpful in future research.

Finally, we use signaling theory to create a link between VC involvement and entrepreneurial exit. Although this theory has begun to receive the interest of scholars in entrepreneurship, the lion's share of the work that has drawn from signaling is still outside the realm of our field. However, there is a substantial opportunity to adopt signaling theory to enhance our understanding of the entrepreneurial cycle. In this paper, our goal was to isolate the partial effects of signals and demonstrate their usefulness in a setting typically characterized by a severe asymmetric information problem. Clearly, the entrepreneurial exit phenomenon is very complex and other theories—such as the resource-based view of the firm, for instance—can and have offered a great deal of insights to help sort out the various determinants of exit. In our view, information economics theory in general and signaling theory in particular can provide a complementary lens through which to examine not only exit, but also other aspects of entrepreneurship. Thus, we hope to inspire future scholarly work that will consider how the inefficiencies brought about by information asymmetry can be mitigated through the use of signals for the benefit of all stakeholders surrounding entrepreneurial companies. Both theoretical and empirical work tackling this question as well as the question of how signaling theory may interact with other well-received theories would be of value to the field.

REFERENCES

Akerlof, G.A. (1970). The market for "lemons": Qualitative uncertainty and the market mechanism. *Quarterly Journal of Economics*, *84*, 488–500.

Aldrich, H.E. (1999). *Organizations evolving*. London: Sage Publications.

Allison, P.D. (1995). The impact of random predictors on comparisons of coefficients between models. *American Journal of Sociology*, *100*, 1294–1305.

Arikan, A.M. & Capron, L. (2010). Dual effects of pre-IPO affiliations on post-IPO acquisition returns. *Strategic Management Journal*, *31*, 1257–1289.

Arrow, K.J. (1974). *The limits of organization*. New York: Norton.

Audretsch, D. & Feldman, M.P. (1996). R&D spillovers and the geography of innovation and production. *American Economic Review*, *86*, 630–640.

Baker, M. & Gompers, P.A. (2003). The determinants of board structure at the initial public offering. *Journal of Law and Economics*, *46*(2), 569–598.

Barzel, Y. (1987). The entrepreneur's reward for self-policing. *Economic Inquiry*, *25*, 103–116.

Bergh, D.D., Connelly, B.L., Ketchen, D.J., & Shannon, L.M. (2014). Signalling theory and equilibrium in strategic management research: An assessment and a research agenda. *Journal of Management Studies*, *41*, 1334–1360.

Brav, A. & Gompers, P.A. (1997). Myth or reality? The long-run underperformance of initial public offerings: Evidence from venture and nonventure capital-backed companies. *Journal of Finance*, *52*, 1791–1821.

Brush, C.G. (1995). *International entrepreneurship: The effect of firm age on motives for internationalization*. New York: Garland Publishing, Inc.

Brush, C.G. & Vanderwerf, P.A. (1992). A comparison of methods and sources for obtaining estimates of new venture performance. *Journal of Business Venturing*, *7*, 157–170.

Capron, L. & Shen, J.-C. (2007). Acquisitions of private vs. public firms: Private information, target selection, and acquirer returns. *Strategic Management Journal*, *28*(9), 891–911.

Carow, K., Heron, R., & Saxton, T. (2004). Do early birds get the returns? An empirical investigation of early-mover advantages with acquisitions. *Strategic Management Journal*, *25*, 563–585.

Carter, R. & Manaster, S. (1990). Initial public offerings and underwriter reputation. *Journal of Finance*, *45*, 1045–1067.

Certo, S.T. (2003). Influencing IPO investors with prestige: Signaling with board structures. *Academy of Management Review*, *28*, 432–446.

Chang, S.J. (2004). Venture capital financing, strategic alliances, and the initial public offerings of Internet startups. *Journal of Business Venturing*, *19*(5), 721–741.

Connelly, B.L., Certo, S.T., Ireland, R.D., & Reutzel, C.R. (2011). Signaling theory: A review and assessment. *Journal of Management*, *37*(1), 39–67.

Coval, J.D. & Moskowitz, T.J. (1999). Home bias at home: Local equity preference in domestic portfolios. *Journal of Finance*, *54*, 2045–2073.

Coval, J.D. & Moskowitz, T.J. (2001). The geography of investment: Informed trading and asset prices. *Journal of Political Economy*, *109*, 811–841.

Covin, J.G. & Slevin, D.P. (1990). New venture strategic posture, structure, and performance: An industry life cycle analysis. *Journal of Business Venturing*, *5*, 123–135.

Cox, D.R. (1972). Regression models and life tables. *Journal of the Royal Statistical Society*, *34*, 1987–1220.

Dushnitsky, G. & Shaver, J.M. (2009). Limitations to inter-organizational knowledge acquisitions: The paradox of corporate venture capital. *Strategic Management Journal*, *30*, 1045–1064.

Elitzir, R. & Gavios, A. (2003). Contracting, signaling, and moral hazard: A model of entrepreneurs, "angels," and venture capitalists. *Journal of Business Venturing*, *18*, 709–725.

Freeman, J.H., Carroll, G.R., & Hannan, M.T. (1983). The liability of newness: Age dependence in organizational death rates. *American Sociological Review*, *48*, 692–710.

Gompers, P. (1996). Grandstanding in the venture capital industry. *Journal of Financial Economics*, *42*, 133–156.

Gompers, P.A. & Lerner, J. (2004). *The venture capital cycle*. Cambridge, MA: MIT Press.

Graebner, M.E. (2009). Caveat venditor: Trust asymmetries in acquisitions of entrepreneurial firms. *Academy of Management Journal*, *52*, 435–472.

Greene, W.H. (1997). *Econometric analysis* (3rd ed.). Upper Saddle River, NJ: Prentice Hall.

Gulati, R. & Higgins, M.C. (2003). Which ties matter when? The contingent effects of interorganizational partnerships on IPO success. *Strategic Management Journal*, *24*, 127–144.

Guler, I. (2007). Throwing good money after bad? Political and institutional influences on sequential decision making in the venture capital industry. *Administrative Science Quarterly*, *52*, 248–285.

Gupta, A.K. & Sapienza, H.J. (1992). Determinants of venture capital firms' preferences regarding the industry diversity and geographic scope of their investments. *Journal of Business Venturing*, *5*, 347–362.

Hagedoorn, J. & Sadowski, B. (1999). The transition from strategic technology alliances to mergers and acquisitions: An exploratory study. *Journal of Management Studies*, *36*, 87–107.

Haveman, H. & Khaire, M. (2004). Survival beyond succession? The contingent impact of founder succession on organizational failure. *Journal of Business Venturing*, *19*, 437–463.

Higgins, M.C. & Gulati, R. (2003). Getting off to a good start: The effects of upper echelon affiliations on underwriter prestige. *Organization Science*, *14*, 244–264.

Higgins, M.C. & Gulati, R. (2006). Stacking the deck: The effects of top management backgrounds on investor decisions. *Strategic Management Journal*, *27*, 1–25.

Hopp, C. & Lukas, C. (2014). A signaling perspective on partner selection in venture capital syndicates. *Entrepreneurship Theory and Practice*, *38*, 635–670.

Hsu, D.H. (2004). What do entrepreneurs pay for venture capital affiliation? *Journal of Finance*, *59*, 1805–1844.

Hsu, D.H. (2006). Venture capitalists and cooperative start-up commercialization strategy. *Management Science*, *52*, 204–221.

Jaffe, A.B., Trajtenberg, M., & Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *Quarterly Journal of Economics*, *63*, 577–598.

Kaplan, S.N. & Strömberg, P. (2003). Financial contracting theory meets the real world: An empirical analysis of venture capital contracts. *Review of Economic Studies*, *70*, 281–315.

Krishnan, C.N.V., Ivanov, V.I., Masulis, R.W., & Singh, A.K. (2011). Venture capital reputation, post-IPO performance, and corporate governance. *Journal of Financial and Quantitative Analysis*, *46*, 1295–1333.

Krishnan, K., Chemmanur, J., & Nandy, D. (2011). How does venture capital financing improve efficiency in private firms? A look beneath the surface. *Review of Financial Studies*, *24*, 4037–4090.

Larsson, R. & Finkelstein, S. (1999). Integrating strategic, organizational, and human resource perspectives on mergers and acquisitions. *Organization Science*, *10*, 1–26.

Lee, C., Lee, K., & Pennings, J.M. (2001). Internal capabilities, external networks, and performance: A study on technology-based ventures. *Strategic Management Journal*, *22*, 615–640.

Lee, P.M. (2001). What's in a name.com? The effects of ".com" name changes on stock prices and trading activity. *Strategic Management Journal*, *22*, 793–804.

Lee, P.M., Pollock, T.G., & Jin, K. (2011). The contingent value of venture capitalist reputation for entrepreneurial firms. *Strategic Organization*, *9*(1), 33–69.

Leiblein, M.J. & Reuer, J.J. (2004). Building a foreign sales base: The roles of capabilities and alliances for entrepreneurial firms. *Journal of Business Venturing*, *19*(2), 285–307.

Lerner, J. (1995). Venture capitalists and the oversight of private firms. *Journal of Finance*, *50*, 301–318.

Lester, R.H., Certo, S.T., Dalton, C.M., Dalton, D.R., & Cannella, A.A. (2006). Initial public offering investor valuations: An examination of top management team prestige and environmental uncertainty. *Journal of Small Business Management*, *44*(1), 1–26.

Manigart, S., Baeyens, K., & Van Hyfte, W. (2002). The survival of venture capital backed companies. *Venture Capital: An International Journal of Entrepreneurial Finance*, *4*(2), 103–124.

Manigart, S. & Wright, M. (2013). Venture capital investors and portfolio firms. *Foundations and Trends in Entrepreneurship*, *9*, 365–570.

Masulis, R.W. & Nahata, R. (2011). Venture capital conflicts of interest: Evidence from acquisitions of venture-backed firms. *Journal of Financial and Quantitative Analysis*, *46*(02), 395–430.

McGrath, R.G. (1999). Falling forward: Real options reasoning and entrepreneurial failure. *Academy of Management Review*, *24*(1), 13–30.

Megginson, W. & Weiss, K. (1991). Venture capitalist certification in initial public offerings. *Journal of Finance*, *46*, 879–903.

Mitchell, W. & Singh, K. (1992). Incumbents' use of pre-entry alliances before expansion into new technical subfields of an industry. *Journal of Economic Behavior and Organization*, *18*, 347–372.

Mody, A. (1993). Learning through alliances. *Journal of Economic Behavior and Organization*, *20*, 151–170.

Nahata, R. (2008). Venture capital reputation and investment performance. *Journal of Financial Economics*, *90*(2), 127–151.

Ozmel, U., Reuer, J.J., & Gulati, R. (2013). Signals across multiple networks: How venture capital and alliance networks affect interorganizational collaboration. *Academy of Management Journal*, *56*(3), 852–866.

Pollock, T.G., Chen, G., Jackson, E.M., & Hambrick, D.C. (2010). How much prestige is enough? Assessing the value of multiple types of high-status affiliates for young firms. *Journal of Business Venturing*, *25*, 6–23.

Pollock, T.G. & Gulati, R. (2007). Standing out from the crowd: The visibility-enhancing effects of IPO-related signals on alliance formation by entrepreneurial firms. *Strategic Organization*, *5*, 339–372.

Porrini, P. (2004). Can a previous alliance between an acquirer and a target affect acquisition performance? *Journal of Management*, *30*, 545–562.

Puranam, P., Powell, B.C., & Singh, H. (2006). Due diligence failure as a signal detection problem. *Strategic Organization*, *4*(4), 319–348.

Puri, M. & Zarutskie, R. (2012). On the life cycle dynamics of venture capital and non venture capital financed firms. *Journal of Finance*, *67*(6), 2247–2293.

Ragozzino, R. & Reuer, J.J. (2007). Initial public offerings and the acquisition of entrepreneurial firms. *Strategic Organization*, *5*, 155–176.

Ragozzino, R. & Reuer, J.J. (2011). Geographic distance and corporate acquisitions: Signals from IPO firms. *Strategic Management Journal*, *32*, 876–894.

Ravenscraft, D.J. & Scherer, F.M. (1987). Life after takeover. *Journal of Industrial Economics*, *2*, 147–156.

Reuer, J.J., & Ragozzino, R. (2012). The choice between joint ventures and acquisitions: Insights from signaling theory. *Organization Science*, *23*(4), 1175–1190.

Reuer, J.J., Tong, T.W., & Wu, C.-W. (2012). A signaling theory of acquisition premiums: Evidence from IPO targets. *Academy of Management Journal*, *55*(3), 667–683.

Roll, R. (1986). The hubris hypothesis of corporate takeovers. *Journal of Business*, *59*, 197–216.

Rosenbusch, N., Brinckmann, J., & Müller, V. (2013). Does acquiring venture capital pay off for the funded firms? A meta-analysis on the relationship between venture capital investment and funded firm financial performance. *Journal of Business Venturing*, *28*(3), 335–353.

Sahlman, W.A. (1990). The structure and governance of venture-capital organizations. *Journal of Financial Economics*, *27*(2), 473–521.

Seth, A. (1990). Sources of value creation in acquisitions: An empirical investigation. *Strategic Management Journal*, *11*, 431–446.

Shane, S. & Cable, D. (2002). Network ties, reputation and the financing of new ventures. *Management Science*, *48*, 364–381.

Shane, S. & Stuart, T. (2002). Organizational endowments and the performance of university start-ups. *Management Science*, *48*, 154–170.

Shrader, R.C. (1996). Influences on and performance implications of internationalization by publicly owned US new ventures: A risk taking perspective. Doctoral dissertation, Georgia State University.

Sørensen, M. (2007). How smart is smart money? A two-sided matching model of venture capital. *Journal of Finance*, *62*, 2725–2762.

Sorenson, O. & Stuart, T.E. (2001). Syndication networks and the spatial distribution of venture capital investments. *American Journal of Sociology*, *106*, 1546–1588.

Spence, M.A. (1973). Job market signaling. *Quarterly Journal of Economics*, *87*, 355–374.

Stuart, T.E., Hoang, H., & Hybels, R.C. (1999). Interorganizational endorsements and the performance of entrepreneurial ventures. *Administrative Science Quarterly*, *44*, 315–349.

Stuart, T.E. & Sorenson, O. (2003). Liquidity events and the geographic distribution of entrepreneurial activity. *Administrative Science Quarterly*, *48*, 175–201.

Wasserman, N. (2003). Founder-CEO succession and the paradox of entrepreneurial success. *Organization Science*, *14*, 149–172.

Zahra, S.A. (1996). Governance, ownership, and corporate entrepreneurship: The moderating impact of industry technological opportunities. *Academy of Management Journal*, *39*, 1713–1735.

Zahra, S.A., Ireland, R.D., & Hitt, M.A. (2000). International expansion by new venture firms: International diversity, mode of market entry, technological learning, and performance. *Academy of Management Journal*, *43*, 925–950.

Zimmerman, M.A. (2008). The influence of top management team heterogeneity on the capital raised through an initial public offering. *Entrepreneurship Theory and Practice*, *32*(3), 391–414.

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[Corrections added on 17 February 2016, after first online publication: the affiliations, telephone number and email addresses of Roberto Ragozzino and Dane P. Blevins have been updated.]